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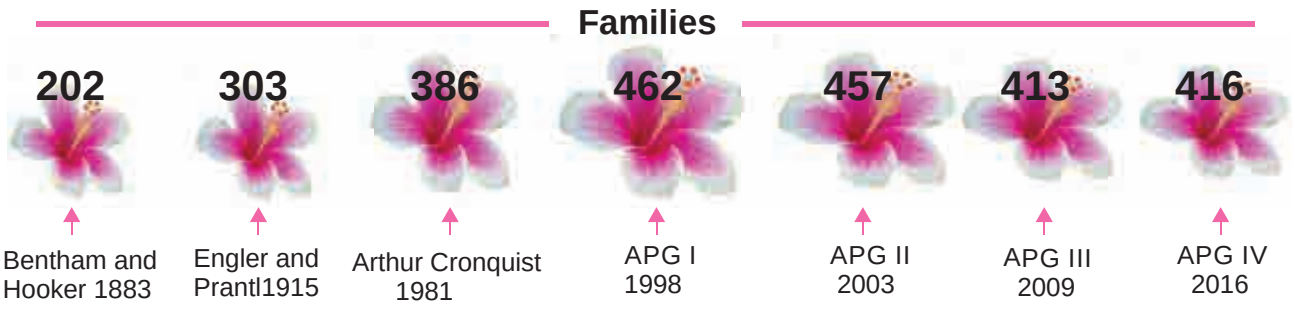


Figure 5.12: A timeline showing the history of classifying flowering plants into families.
(Source: Royal Botanic Gardens Kew State of World's Plant 2017)

Classification reflects the state of our knowledge at a given point of time. It will continue to change as we acquire new information.



A significant number of major herbaria, including Kew are changing the order of their collections in accordance with APG.

The influential world checklist of selected plant families (also from kew) is being updated to the APG III system.

A recent photographic survey of the plants of USA and Canada is organized according to the APG III system.

In UK, the latest edition of the standard flora of the British Isles written by Stace is based on the APG III system.

5.11 Modern trends in taxonomy

Taxonomists now accept that, the morphological characters alone should not be considered in systematic classification of plants. The complete knowledge of taxonomy is possible with the principles of various disciplines like Cytology, Genetics, Anatomy, Physiology, Geographical Distribution, Embryology, Ecology, Palynology, Phenology, Bio-Chemistry, Numerical Taxonomy and Transplant Experiments. These have been found to be useful in solving some of the taxonomical problems by providing additional characters. It has changed the face of classification from **alpha** (classical) to **omega** (modern kind). Thus the new systematic has evolved into a better taxonomy.

5.11.1 Chemotaxonomy

Various medicines, spices and preservatives obtained from plant have drawn the attention of Taxonomists. Study of various chemicals available in plants help to solve certain taxonomical problems. Chemotaxonomy is the scientific approach of classification of plants on the basis of their biochemical constituents. As proteins are more closely controlled by genes and less subjected to natural selection, it has been used at all hierarchical levels of classification starting from the rank of 'variety' up to the rank of division in plants. Proteins, amino acids, nucleic acids, peptides etc. are the most studied chemicals in chemotaxonomy.

The chemical characters can be divided into three main categories.